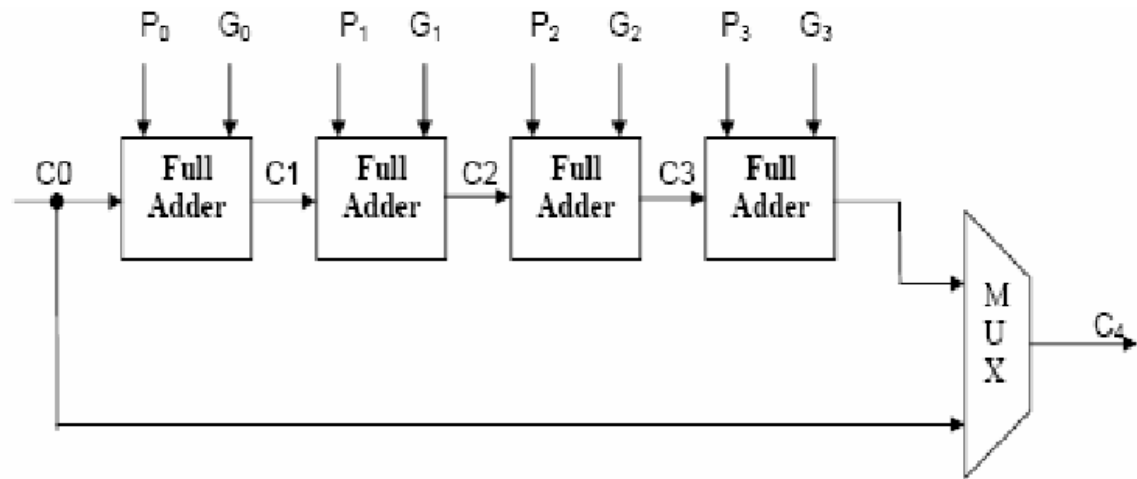


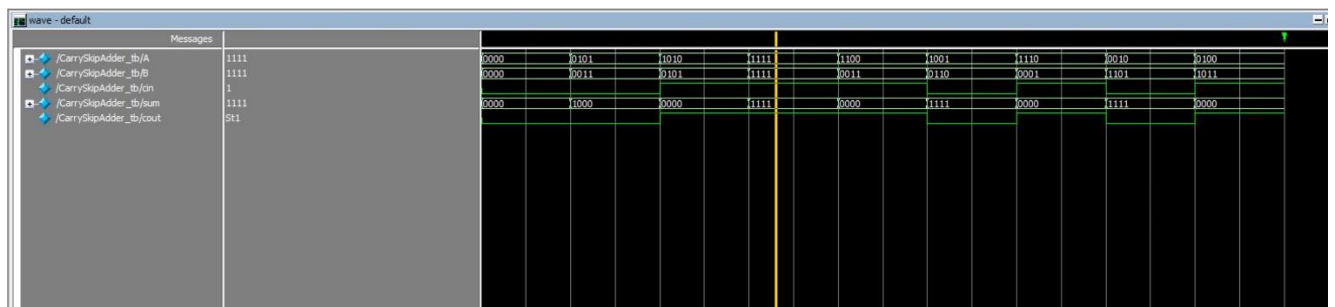
## HW1

Implement a carry skip adder.

Carry skip adder schema:



Full wave:



Code:

```

module FA(
    input  a, b, cin,
    output sum, cout
);
    assign {cout, sum} = a + b + cin;
endmodule

module mux2to1(
    input  a, b, sel,
    output y
);
    assign y = sel ? a : b;
endmodule

module CarrySkipAdder (
    input  [3:0] A, B,
    input  cin,
    output [3:0] sum,
    output cout
);
    wire [3:0] c;
    wire [3:0] p;
    wire P_sel;

    FA fa0 (.a(A[0]), .b(B[0]), .cin(cin), .sum(sum[0]), .cout(c[0]));
    FA fa1 (.a(A[1]), .b(B[1]), .cin(c[0]), .sum(sum[1]), .cout(c[1]));
    FA fa2 (.a(A[2]), .b(B[2]), .cin(c[1]), .sum(sum[2]), .cout(c[2]));
    FA fa3 (.a(A[3]), .b(B[3]), .cin(c[2]), .sum(sum[3]), .cout(c[3]));

    assign p[0] = A[0] ^ B[0];
    assign p[1] = A[1] ^ B[1];
    assign p[2] = A[2] ^ B[2];
    assign p[3] = A[3] ^ B[3];

    assign P_sel = p[0] & p[1] & p[2] & p[3];

    mux2to1 mux_inst (.a(c[3]), .b(cin), .sel(P_sel), .y(cout));
endmodule

module CarrySkipAdder_tb;

    reg [3:0] A, B;
    reg cin;
    wire [3:0] sum;
    wire cout;

    CarrySkipAdder dut (.A(A), .B(B), .cin(cin), .sum(sum), .cout(cout));

    initial begin
        A=4'b0000; B=4'b0000; cin=0; #10;
        A=4'b0101; B=4'b0011; cin=0; #10;
        A=4'b1010; B=4'b0101; cin=1; #10;
        A=4'b1111; B=4'b1111; cin=1; #10;
        A=4'b1100; B=4'b0011; cin=1; #10;
        A=4'b1001; B=4'b0110; cin=0; #10;
        A=4'b1110; B=4'b0001; cin=1; #10;
        A=4'b0010; B=4'b1101; cin=0; #10;
        A=4'b0100; B=4'b1011; cin=1; #10;

        $finish;
    end
endmodule

```

Wave:

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 0000 | 0000 |
| + /CarrySkipAdder_tb/B   | 0000 | 0000 |
| /CarrySkipAdder_tb/cin   | 0    |      |
| + /CarrySkipAdder_tb/sum | 0000 | 0000 |
| /CarrySkipAdder_tb/cout  | St0  |      |

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 0101 | 0101 |
| + /CarrySkipAdder_tb/B   | 0011 | 0011 |
| /CarrySkipAdder_tb/cin   | 0    |      |
| + /CarrySkipAdder_tb/sum | 1000 | 1000 |
| /CarrySkipAdder_tb/cout  | St0  |      |

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 1010 | 1010 |
| + /CarrySkipAdder_tb/B   | 0101 | 0101 |
| /CarrySkipAdder_tb/cin   | 1    |      |
| + /CarrySkipAdder_tb/sum | 0000 | 0000 |
| /CarrySkipAdder_tb/cout  | St1  |      |

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 1111 | 1111 |
| + /CarrySkipAdder_tb/B   | 1111 | 1111 |
| /CarrySkipAdder_tb/cin   | 1    |      |
| + /CarrySkipAdder_tb/sum | 1111 | 1111 |
| /CarrySkipAdder_tb/cout  | St1  |      |

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 1100 | 1100 |
| + /CarrySkipAdder_tb/B   | 0011 | 0011 |
| /CarrySkipAdder_tb/cin   | 1    |      |
| + /CarrySkipAdder_tb/sum | 0000 | 0000 |
| /CarrySkipAdder_tb/cout  | St1  |      |

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 1001 | 1001 |
| + /CarrySkipAdder_tb/B   | 0110 | 0110 |
| /CarrySkipAdder_tb/cin   | 0    |      |
| + /CarrySkipAdder_tb/sum | 1111 | 1111 |
| /CarrySkipAdder_tb/cout  | St0  |      |

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 1110 | 1110 |
| + /CarrySkipAdder_tb/B   | 0001 | 0001 |
| /CarrySkipAdder_tb/cin   | 1    |      |
| + /CarrySkipAdder_tb/sum | 0000 | 0000 |
| /CarrySkipAdder_tb/cout  | St1  |      |

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 0010 | 0010 |
| + /CarrySkipAdder_tb/B   | 1101 | 1101 |
| /CarrySkipAdder_tb/cin   | 0    |      |
| + /CarrySkipAdder_tb/sum | 1111 | 1111 |
| /CarrySkipAdder_tb/cout  | St0  |      |

| wave - default           |      |      |
|--------------------------|------|------|
| Messages                 |      |      |
| + /CarrySkipAdder_tb/A   | 0100 | 0100 |
| + /CarrySkipAdder_tb/B   | 1011 | 1011 |
| /CarrySkipAdder_tb/cin   | 1    |      |
| + /CarrySkipAdder_tb/sum | 0000 | 0000 |
| /CarrySkipAdder_tb/cout  | St1  |      |